



Decoding Skewness: Understanding Data Distribution

Section 1: Learn

What is Skewness in Data?

Skewness describes the **asymmetry** in a dataset's distribution.

A dataset is **symmetrical** if it looks the same on both sides of its mean.

Why is Skewness Important?

- **Identifies Data Imbalance** – Helps understand **how data is distributed**.
- **Impacts Statistical Analysis** – Many tests **assume normal distribution**.
- **Improves Decision-Making** – Skewed data affects **business and research insights**.

Types of Skewness

1. **Symmetrical (No Skewness)** – The left and right sides of the data distribution are **equal**.
2. **Positively Skewed (Right Skewed)** – Tail is longer on the right, meaning a few very high values.
 - *Example:* Income distribution (few people earn extremely high salaries).
3. **Negatively Skewed (Left Skewed)** – Tail is longer on the left, meaning a few very low values.
 - *Example:* Exam scores (most students score high, few score very low).

How Do We Measure Skewness?

- **Skewness Coefficient (γ_1)**



- $\gamma_1 = 0 \rightarrow$ Symmetrical
- $\gamma_1 > 0 \rightarrow$ Right Skewed
- $\gamma_1 < 0 \rightarrow$ Left Skewed
- **Mean, Median, and Mode Relationship**
 - **Symmetrical:** Mean = Median = Mode
 - **Right Skewed:** Mean > Median > Mode
 - **Left Skewed:** Mean < Median < Mode

Anecdote: How Skewness Helped in Salary Analysis

A company wanted to determine the **average salary** of employees.

Using the **mean salary**, they got ₹1,20,000, but most employees actually earned around ₹50,000.

The data was **right-skewed** due to a few **high-earning executives**.

By using the **median instead of the mean**, they got a more **accurate picture** of typical salaries.

Section 2: Practice

1. Identifying Skewness in a Dataset

Example: Monthly Salaries of 7 Employees

Employee	Salary (₹)
A	30,000
B	35,000
C	40,000
D	50,000



Employee	Salary (₹)
E	55,000
F	1,50,000
G	2,00,000

- **Mean** = ₹80,000 (pulled up by high salaries)
 - **Median** = ₹50,000 (better represents typical earnings)
 - **Skewness:** Right-Skewed (few high values pull the mean upward).
-

2. Understanding Left Skewness

Example: Exam Scores of 6 Students

Student	Score
A	85
B	88
C	90
D	92
E	95
F	40

- **Mean** = 81.67
 - **Median** = 89
 - **Skewness:** Left-Skewed (one very low score pulls the mean downward).
-



3. Using Mean, Median, and Mode for Analysis

Dataset Type	Mean vs. Median
Symmetrical	Mean $\approx$ Median $\approx$ Mode
Right Skewed	Mean $>$ Median
Left Skewed	Mean $<$ Median

Section 3: Know More

Frequently Asked Questions (FAQs)

1. What does a skewness value of 0 mean?

It means the dataset is **perfectly symmetrical**, with **equal distribution** on both sides.

2. How does skewness affect data interpretation?

If data is skewed, using the **mean alone** can be misleading. The **median is often a better measure**.

3. What is the difference between positive and negative skewness?

- **Positive Skewness** → Data has **high extreme values**, pulling the mean up.
- **Negative Skewness** → Data has **low extreme values**, pulling the mean down.



4. How does skewness impact machine learning?

Many machine learning algorithms **assume normal distribution**. Highly skewed data may **affect accuracy** and require **transformations**.

5. How can I reduce skewness in data?

- **Log Transformation** – Helps in **right-skewed** data.
- **Square Root Transformation** – Reduces the effect of large values.

Conclusion:

Understanding skewness helps in **interpreting real-world data accurately**. By analyzing the **shape of distribution, mean-median relation, and outliers**, we can **make better decisions in business, research, and data science**.